

Ziqi Deng

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Education

B.Eng. Harbin Institute of Technology, Weihai (HITWH), Electromagnetic Fields and Wireless Technology Sept 2020 – June 2024

- GPA: 80.59/100 | **Outstanding Graduate**
- IELTS: 6.0

Professional Experience

Huawei Technologies Co., Ltd., Embedded Software Engineer (Level B) July 2024 – present

- **FDD-RRU Decoupled Software Development (MDE)**: Acting as a technical lead for the Decoupling Competence Center. Orchestrated software maintenance for full-series FDD-RRU production lines. Diagnosed and resolved **50+ complex hardware-software anomalies**, ensuring **zero downtime** implemented white-box testing to preemptively intercept **10+ potential risks**.
- **RRU Business Logic & Feature Delivery (FO)**: Spearheaded the end-to-end delivery of **20+ system requirements**, including complex tasks for Tier-1 operators (e.g., DTP, AHCS Alarms). Contributed **20k+ lines** of high-quality C/C++ code, ensuring **zero defects** in backend acceptance tests.
- **Full-lifecycle Module Development**: Led the delivery of 2 new-gen RRUs and participated in 7 others. Realized the industry's first RU product supporting **N75 to B32 frequency band soft-switching**, enabling the first commercial deployment of Huawei's AHCS features.

Shandong Platinum Power Technology Co., Ltd., Embedded Software Engineering Intern Jan 2023 – May 2024

- **MCU Firmware & RTOS Adaptation**: Developed cross-platform firmware based on **RT-Thread** for STM32/GD32. Implemented an **Adaptive Communication Protocol** supporting CAN, SPI, and IIC for underwater robotic sensors.
- **PCB Design & PCBA Prototyping**: Designed **2/4-layer PCBs** using Altium Designer. Managed hardware prototyping and signal integrity debugging for robot logging systems and depth sensors.
- **Full-cycle R&D & Tech-Transfer**: Managed the complete project lifecycle and authored **5 National Invention Patents** and **1 Utility Model Patent** (All Granted).

Project Experience

Full-stack Robotics Logging System & IPC Framework (RT-Thread) Project Lead Oct 2022 – Apr 2023

- **Middleware Architecture**: Designed a lightweight **Publish-Subscribe communication middleware** on RT-Thread, achieving architectural decoupling between application logic and hardware abstraction layers (HAL).
- **Concurrency & Fault Diagnosis**: Implemented mutex protection and asynchronous logging to resolve data race conditions. Established an offline fault diagnosis scheme utilizing the **FatFS** file system for tiered storage.
- **System Integration**: Engineered a custom PCBA (STM32F407ZET6) to support the framework, aiding the team in pinpointing sporadic system crashes during the 2023 season. Core technology protected by **6 granted patents**.

Embedded Control Systems for RoboMaster 2022 Sentry & Dart Robots Embedded System Lead Dec 2021 – Sept 2022

- **Sentry Robot**: Optimized MCU-Vision communication protocols to eliminate computational bottlenecks. Developed a **Randomized Motion Strategy** based on RNG peripherals and a Finite State Machine (FSM) for evasive maneuvers, securing a 'Triple Kill' in regional competitions.
- **Dart Robot**: Engineered an **Adaptive Ballistic Correction** algorithm using closed-loop correction coefficients, replacing manual alignment with an automated, high-precision targeting system.

Non-preview Electromagnetic Racing Car (Smart Car Competition)

- Optimized **Servo PD Control Algorithms** for **Infineon TriCore (TC264/TC364)** multi-core architecture, achieving high-speed stability under extreme non-preview sensor constraints.
- Designed heuristic recovery algorithms for complex track elements (e.g., right-angle turns, bumps). Awarded **Provincial 2nd Prize** in the Shandong Division.

Embedded Lead
Dec 2020 – Aug 2021

Campus Leadership

HERO Robotics Team (Deputy Lead of Embedded Group / Consultant)

Dec 2022 – May 2024

- Responsible for the full-cycle development of embedded systems for Sentry, UAV, and Dart robot platforms.
- Standardization & Mentorship:** Authored over **80,000 words** of technical documentation and learning modules. Led the team's architectural migration from bare-metal to **RT-Thread RTOS**.
- Knowledge Management:** Established standardized communication interfaces and low-level code repositories, significantly reducing the ramp-up time for new team members.

Patent Achievements

Method for obtaining operating information of embedded systems and computer-readable storage medium (Invention Patent)

2023

Ziqi Deng (1st Author)

ZL202310560683.5 (Granted)

A control method for embedded operating systems (Invention Patent)

2023

Ziqi Deng (2nd Author)

ZL202310561393.2 (Granted)

A hardware-decoupled multi-address variable management method (Invention Patent)

2023

Ziqi Deng (Student 1st Author)

ZL202310647052.7 (Granted)

A monitoring method for embedded robot operation status (Invention Patent)

2023

Ziqi Deng (Student 2nd Author)

ZL202310647080.9 (Granted)

A log recording method and log recording device (Invention Patent)

2023

Ziqi Deng (Student 1st Author)

ZL202310647068.8 (Granted)

Honors and Awards

- National 2nd Prize** | 22nd RoboMaster University Championship, National Final (2023)
- National 2nd Prize** | 21st RoboMaster University Championship, National Final (2022)
- National 2nd Prize** | RoboMaster Specialized Combat Award (Sentry Robot) (2022)
- 1st Prize** | RoboMaster University Championship, Central/Eastern Regional (2022, 2023)
- Provincial 2nd Prize** | 16th National University Smart Car Competition, Shandong Division (2021)
- Outstanding Graduate** | Harbin Institute of Technology, Weihai (2024)
- Academic Scholarships:** 2nd Class People's Scholarship (Top 10%, 2 times) & Social Work Scholarship (5 times)

Skills

Embedded Systems: C/C++, RT-Thread RTOS, uORB Architecture, TriCore (Infineon), STM32/GD32, Driver Development

Hardware Design: Altium Designer, 2/4-layer PCB Design, PCBA Debugging, Communication Protocols (CAN/SPI/IIC/RS485)

Engineering Tools: Git, Technical Documentation (80k+ words), Fault Diagnosis, White-box Testing